

$$2) \text{ Formula: } \frac{M_{eq} + M_0}{M_{eq}} = \left(\frac{f'}{f''} \right)^2 \quad \frac{M_{eq} + 0.1}{M_{eq}} = \left(\frac{11.396}{10.840} \right)^2$$

$$M_{eq} + 0.1 = 1.1052 M_{eq}$$

$$0.1052 M_{eq} = 0.1$$

i) Equivalent mass of the horizontal vibration table, $M_{eq} = \underline{0.9506 \text{ kg}}$

0.6 A

ii) System's frequency, f_d

$$T_d = \frac{\Delta t}{4} = \frac{0.578 - 0.234}{4} = 0.086 \quad f_d = \frac{1}{T_d} = \frac{1}{0.086} = \underline{11.6279 \text{ Hz}}$$

$$\text{iii) Damping ratio } \xi = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} \quad \delta = \frac{1}{4} \ln \left(\frac{x_i}{x_{i+4}} \right) = n T_d$$

$$\delta = \frac{1}{4} \ln \left(\frac{0.3923}{0.0857} \right) = 0.3803$$

$$\xi = \frac{0.3803}{\sqrt{4\pi^2 + 0.3803^2}} = \underline{0.0604}$$

1.0 A

$$\text{ii) } T_d = \frac{0.496 - 0.152}{4} = 0.086 \quad f_d = \underline{11.6279 \text{ Hz}}$$

$$\text{iii) } \delta = \frac{1}{4} \ln \left(\frac{0.5413}{0.0678} \right) = 0.5194 \quad \xi = \underline{0.0824}$$

0.8 A

$$\text{ii) } T_d = \frac{0.515 - 0.171}{4} = 0.086 \quad f_d = \underline{11.6279 \text{ Hz}}$$

$$\text{iii) } \delta = \frac{1}{4} \ln \left(\frac{0.4358}{0.078} \right) = 0.4301 \quad \xi = \underline{0.0683}$$

3) Amplitude - Frequency Characteristics

The greater the amperage, the lower the biggest B_m , the greater the damping ratio hence greater the damping effect.

0.6 A

$$B_m = 2.111 \quad f_m = 11.23 \text{ Hz}$$

$$f_n \approx f_m = 11.23 \text{ Hz}$$

$$B_0 = 0.056 \quad f_0 = 3 \text{ Hz}$$

$$\xi \approx \frac{f_2 - f_1}{2f_n} = \frac{11.81 - 10.72}{2(11.23)} = \underline{0.0488}$$

1.0 A

$$B_m = 1.263 \quad f_m = 11.36 \text{ Hz}$$

$$f_n \approx f_m = 11.36 \text{ Hz}$$

$$B_0 = 0.056 \quad f_0 = 3 \text{ Hz}$$

$$\xi = \frac{f_2 - f_1}{2f_n} = \frac{12.36 - 10.81}{2(11.36)} = \underline{0.0814}$$

0.8 A

$$B_m = 1.559 \quad f_m = 11.26 \text{ Hz}$$

$$f_n \approx f_m = 11.26 \text{ Hz}$$

$$B_0 = 0.056 \quad f_0 = 3 \text{ Hz}$$

$$\xi = \frac{f_2 - f_1}{2f_n} = \frac{12.17 - 10.64}{2(11.26)} = \underline{0.0679}$$

4) Phase - Frequency Characteristics

The greater the amperage, the higher the excitation frequency, f_m .

0.6 A

f near the phase resonance point $\approx f_n$

$$f_n = 11.23 \text{ Hz}$$

$$\xi = \frac{\Delta \phi}{f_n \Delta \phi_x} = \frac{11.25 - 11.19}{11.23 \left(\frac{52}{180} \pi \right)} = \underline{0.0589}$$

1.0 A

$$f_n = 11.35 \text{ Hz}$$

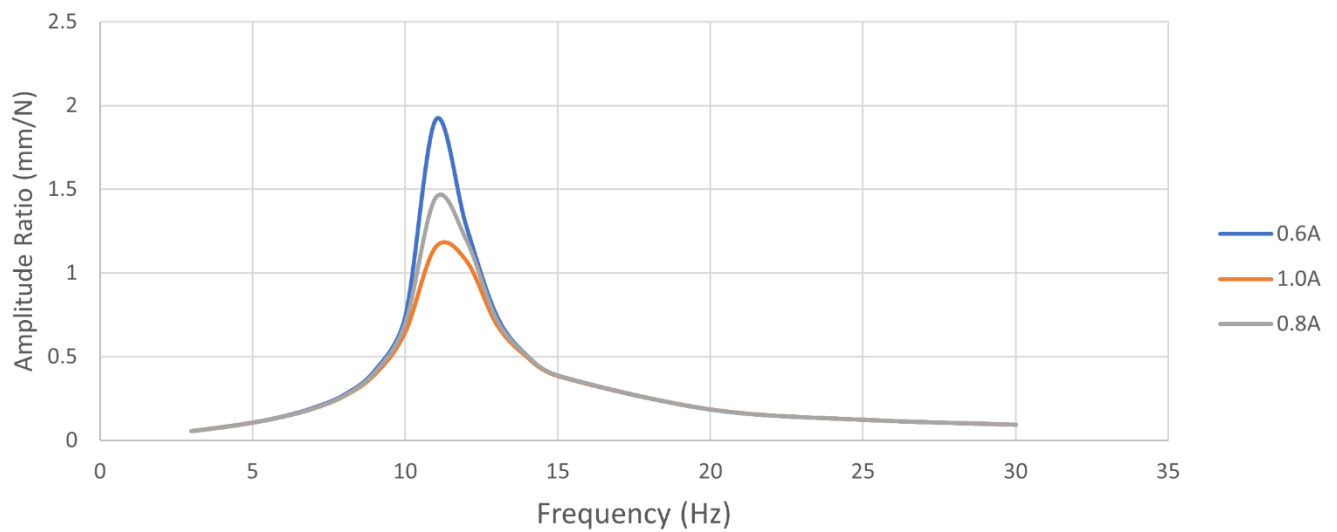
$$\xi = \frac{\Delta \phi}{f_n \Delta \phi_x} = \frac{11.38 - 11.30}{11.35 \left(\frac{54}{180} \pi \right)} = \underline{0.0748}$$

$$\underline{0.8A}$$

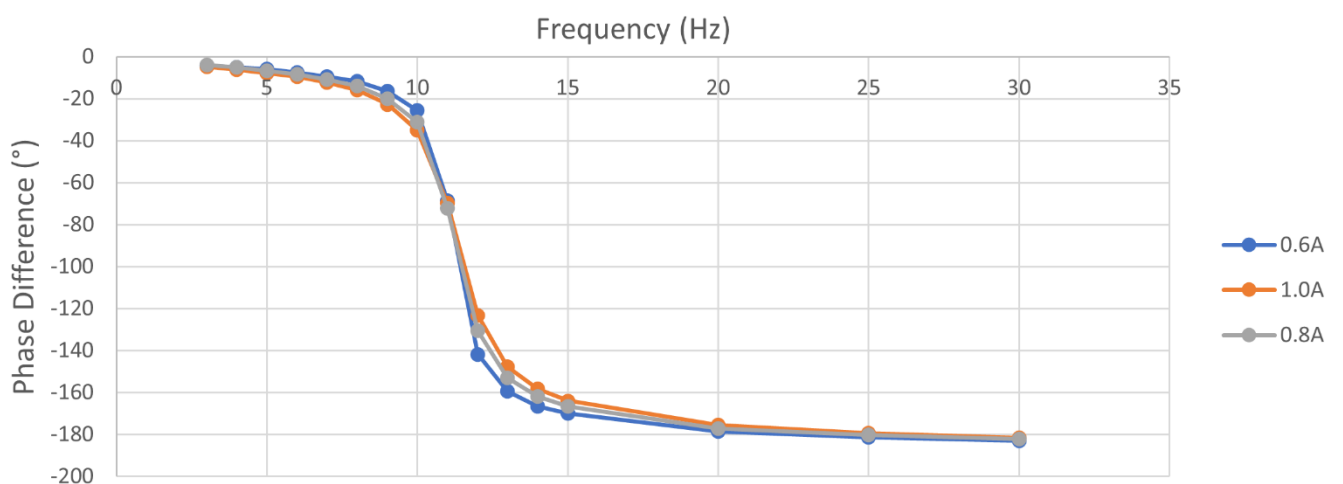
$$f_n = 11.25 \text{ Hz}$$

$$\xi = \frac{\Delta f}{f_n \gamma_x} = \frac{11.30 - 11.20}{11.25 \left(\frac{57}{180} \pi \right)} = \underline{0.0894}$$

The curve of Amplitude-Frequency Characteristics



The curve of Phase-Frequency Characteristics



5) i) Damped vibration method is the most sensitive to damping effects and Phase-frequency method is the least sensitive to damping effects as damping information is inferred indirectly from phase shifts. That explains why the damping ratio obtained from formula

$$\xi = \frac{\Delta f}{f_n \Delta \phi_x} \quad \Delta \phi_x = \text{phase change (rad)}$$

is not relatively close to the ratios obtained by other methods.

ii) As for applicability, amplitude-frequency method is applicable for harmonic excitations and resonant systems. Phase-frequency method is suitable for non-harmonic excitations and systems with varying phase responses. Lastly, damped vibration method is applicable for observing the system's natural response to initial excitation and decay.

iii) Damping ratios obtained by damped vibration and amplitude-frequency method increases with current, indicating a stronger damping effect at higher amperages.